

Readify: An Attention Based Personalized Book Recommendation System

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Abstract

This project proposes **Readify**, an explainable attention-based book recommendation system using Graph Neural Networks (GNN). The system represents users, books, reviews, and interactions as a graph structure and applies Graph Attention Networks (GAT) to learn the importance of different relationships dynamically. Unlike traditional recommendation methods where all interactions are treated equally, Readify assigns attention weights to user behavior, reviews, and connections to identify the most influential factors behind recommendations. The proposed system aims to improve personalized recommendation performance while providing explanations about why a particular book was recommended.

1 Introduction

Recommendation systems are widely used in modern platforms such as online bookstores, streaming services, and e-commerce applications. These systems analyze user behavior to suggest relevant items.

Existing recommendation approaches mainly include:

- Collaborative filtering
- Content-based filtering
- Matrix factorization techniques

However, these approaches have limitations:

- They treat user interactions equally.
- They ignore relationships between users.
- They cannot effectively utilize review influence.
- They provide limited explanation for recommendations.

Readify addresses these problems by modeling recommendation as a graph learning problem where attention mechanisms determine the importance of different relationships.

2 Problem Statement

Given a collection of users, books, reviews, and interactions, the objective of Readify is to predict which books a user is likely to prefer.

3 Objectives

The objectives of this project are:

- To design a attention based representation of users, books, and reviews.
- To implement an attention based recommendation model using Graph Neural Networks.
- To learn dynamic importance weights for user interactions.
- To provide explainable recommendations using attention scores.
- To compare the proposed model with traditional recommendation approaches.

4 Proposed System

The recommendation problem is represented as a heterogeneous graph:

$$G = (V, E)$$

where:

- V represents nodes
- E represents relationships

The graph contains:

- User nodes
- Book nodes
- Review nodes
- User-user relationships
- User-book interaction edges

Example:

$$User_A \rightarrow Review \rightarrow Book_X$$

$$User_A \rightarrow User_B$$

This graph structure allows the system to learn both direct preferences and social influence.

5 Project Contribution

The proposed project contribution includes:

5.1 Review-Aware Attention

The system considers reviews as important information sources instead of only using ratings.

The model learns:

- Which reviews are influential
- Which users provide reliable recommendations
- Which interactions affect user preference

5.2 Dynamic User Influence Learning

Different users influence recommendations differently.

The attention mechanism automatically learns:

- Similar user importance
- Social relationship strength
- Preference similarity

5.3 Explainable Recommendation

Instead of only recommending books, Readify provides reasoning:

Example:

”This book was recommended because users with similar interests rated it highly and your previous reading history matches this genre.”

6 System Architecture

The proposed workflow:

1. Collect book, user, and review dataset.
2. Construct user-book-review graph.
3. Generate node feature representations.
4. Apply Graph Attention Network layers.
5. Learn attention weights.
6. Calculate recommendation scores.
7. Recommend top-k books.

7 Implementation Plan

Algorithm 1 Readify Recommendation Algorithm

Load dataset
Create user-book-review graph
Initialize node embeddings
Apply Graph Attention layers
Calculate attention scores
Generate user preference vector
Rank candidate books
Recommend top-k books

8 Expected Outcome

The expected outcomes are:

- Improved personalized recommendations
- Better understanding of user preferences
- Attention-based explanation of recommendations
- Improved utilization of social and review information

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